File Edit Selection View Go Run Terminal Help

503Q-lab7

EXPLORER

- 503Q-LAB7
  - .terraform
  - userdata
    - backend.sh
    - web.sh
  - .gitignore
  - .terraform.loc... U
  - alb.tf
  - compute.tf
  - main.tf
  - network.tf
  - outputs.tf
  - secrets.tf
  - security.tf
  - terraform.tfstate
  - terraform.tfvars
  - variables.tf

variables.tf | terraform.tfvars | .terraform.lock.hcl U | secrets.tf | outputs.tf

terraform.tfvars > aws_region

```
key_name = "my-ec2"
my_ip = "185.132.132.57/32"
aws_region = "us-east-1"
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL AZURE

```
aws_nat_gateway.main: Still creating... [00m20s elapsed]
aws_autoscaling_group.backend: Still creating... [00m10s elapsed]
aws_autoscaling_group.web: Creation complete after 17s [id=lab007-web-asg]
aws_autoscaling_group.backend: Creation complete after 14s [id=lab007-backend-asg]
aws_lb.external: Still creating... [00m30s elapsed]
aws_nat_gateway.main: Still creating... [00m30s elapsed]
aws_lb.external: Still creating... [00m40s elapsed]
aws_nat_gateway.main: Still creating... [00m40s elapsed]
aws_lb.external: Still creating... [00m50s elapsed]
aws_nat_gateway.main: Still creating... [00m50s elapsed]
aws_lb.external: Still creating... [01m00s elapsed]
aws_nat_gateway.main: Still creating... [01m00s elapsed]
aws_lb.external: Still creating... [01m10s elapsed]
aws_nat_gateway.main: Still creating... [01m10s elapsed]
aws_lb.external: Still creating... [01m20s elapsed]
aws_nat_gateway.main: Still creating... [01m20s elapsed]
aws_lb.external: Still creating... [01m30s elapsed]
aws_nat_gateway.main: Still creating... [01m30s elapsed]
aws_lb.external: Still creating... [01m40s elapsed]
aws_nat_gateway.main: Still creating... [01m40s elapsed]
aws_lb.external: Still creating... [01m50s elapsed]
aws_nat_gateway.main: Still creating... [01m50s elapsed]
aws_nat_gateway.main: Still creating... [02m00s elapsed]
aws_lb.external: Still creating... [02m00s elapsed]
aws_nat_gateway.main: Creation complete after 2m3s [id=nat-028ef34c7a19449df]
aws_route_table.private: Creating...
aws_route_table.private: Creation complete after 3s [id=rtb-0de44e6926f78ff18]
aws_route_table_association.private_1: Creating...
aws_route_table_association.private_2: Creating...
aws_route_table_association.private_2: Creation complete after 1s [id=rtbassoc-0f73a81df6fa37a01]
aws_route_table_association.private_1: Creation complete after 2s [id=rtbassoc-0bcc973e95aae5178]
aws_lb.external: Still creating... [02m10s elapsed]
aws_lb.external: Creation complete after 2m18s [id=arn:aws:elasticloadbalancing:us-east-1:838054321169:loadbalancer/app/lab007-external-alb/f18929a54268ab48]
aws_lb_listener.http: Creating...
aws_lb_listener.http: Creation complete after 2s [id=arn:aws:elasticloadbalancing:us-east-1:838054321169:listener/app/lab007-external-alb/f18929a54268ab48/a8070fcc4b5c96c1]

Apply complete! Resources: 32 added, 0 changed, 0 destroyed.

Outputs:

alb_dns_name = "lab007-external-alb-2146519376.us-east-1.elb.amazonaws.com"
bastion_public_ip = "98.88.200.289"
ssm_parameter_paths = [
  "/lab007/db_password",
  "/lab007/app_env",
]
vpc_id = "vpc-004481cbe10308ffc"
(base) PS C:\Users\royba\Downloads\503Q-lab7>
```

powershell
powershell

OUTLINE
TIMELINE

main* ⊗ 0 ⚠ 0 [Azurite Table Service] [Azurite Queue Service] [Azurite Blob Service] Ln 3, Col 25 Spaces: 4 UTF-8 CRLF terraform-vars Go Live Prettier

Inbox - Roy Baroudy (Stu | Step-by-step code explan | Course: EECE503Q - STD | Koh Lanta S03E02 - You | baroude | IAM | Global | Instances | EC2 | us-east-1 | Two-Tier App

VPN Not secure lab007-external-alb-2146519376.us-east-1.elb.amazonaws.com

# Two-Tier App — Roy Baroudy and Marc Abou Nader

Web Tier → Nginx Reverse Proxy → Backend API

## Backend Health Check

healthy

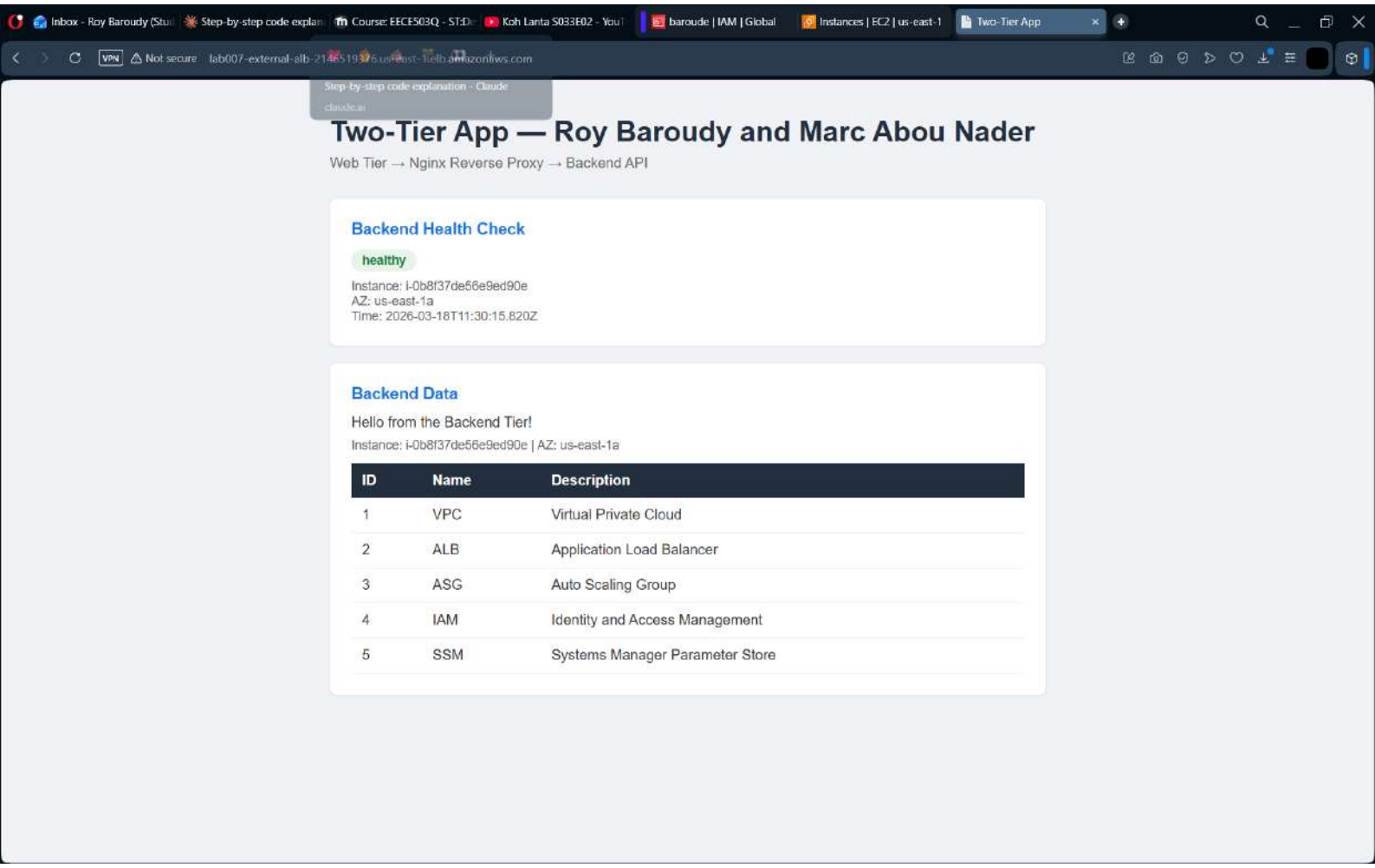
Instance: i-08da3d61cc0682b88
AZ: us-east-1b
Time: 2026-03-18T11:22:59.370Z

## Backend Data

Hello from the Backend Tier!

Instance: i-08da3d61cc0682b88 | AZ: us-east-1b

| ID | Name | Description |
|---|---|---|
| 1 | VPC | Virtual Private Cloud |
| 2 | ALB | Application Load Balancer |
| 3 | ASG | Auto Scaling Group |
| 4 | IAM | Identity and Access Management |
| 5 | SSM | Systems Manager Parameter Store |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Step-by-step code explan
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baroude | IAM | Global
Instances | EC2 | us-east-1
Two-Tier App
VPN
Not secure
lab007-external-alb-2146519376.us-east-1elb.amazonaws.com
Step-by-step code explanation - Claude
claude.ai
Two-Tier App — Roy Baroudy and Marc Abou Nader
Web Tier → Nginx Reverse Proxy → Backend API
Backend Health Check
healthy
Instance: i-0b8f37de56e9ed90e
AZ: us-east-1a
Time: 2026-03-18T11:30:15.820Z
Backend Data
Hello from the Backend Tier!
Instance: i-0b8f37de56e9ed90e | AZ: us-east-1a
ID Name Description
1 VPC Virtual Private Cloud
2 ALB Application Load Balancer
3 ASG Auto Scaling Group
4 IAM Identity and Access Management
5 SSM Systems Manager Parameter Store

Inbox - Roy Baroudy (Stu | Step-by-step code explan | Course: EECE503Q - STD | Koh Lanta S03E02 - YouT | baroude | IAM | Global | Instances | EC2 | us-east-1 | Two-Tier App

us-east-1.console.aws.amazon.com/ec2/home

aws | Search [Alt+S] | United States (N. Virginia) | RoyBaroudy (8380-5432-1169) | RoyBaroudy

EC2 > Instances

**EC2**

- Dashboard
- AWS Global View
- Events
- **Instances**
  - Instances
  - Instance Types
  - Launch Templates
  - Spot Requests
  - Savings Plans
  - Reserved Instances
  - Dedicated Hosts
  - Capacity Reservations
  - Capacity Manager New
- **Images**
  - AMIs
  - AMI Catalog
- **Elastic Block Store**
  - Volumes
  - Snapshots
  - Lifecycle Manager
- **Network & Security**
  - Security Groups
  - Elastic IPs

## Instances (1/5) Info

Connect | Instance state | Actions | Launch instances

Find Instance by attribute or tag (case-sensitive) | All states | 1

| Name | Instance ID | Instance state | Instance type | Status check | Alarm status | Availability Zone | Public IPv4 |
|---|---|---|---|---|---|---|---|
| Bastion-Host | i-0e8c416ca450d3775 | Running | t3.micro | 3/3 checks passed | View alarms + | us-east-1a | ec2-98-80-2 |
| lab007-web | i-0a8df5f86a11d9169 | Running | t3.micro | 3/3 checks passed | View alarms + | us-east-1b | ec2-54-208 |
| lab007-backend | i-08da3d61cc0682b88 | Running | t3.micro | 3/3 checks passed | View alarms + | us-east-1b | – |
| lab007-backend | i-0b8f37de56e9ed90e | Running | t3.micro | 3/3 checks passed | View alarms + | us-east-1a | – |
| lab007-web | i-01c0d55c428585968 | Running | t3.micro | 3/3 checks passed | View alarms + | us-east-1a | ec2-100-26 |

## i-0b8f37de56e9ed90e (lab007-backend)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

### Instance summary Info

| | | |
|---|---|---|
| **Instance ID**<br>i-0b8f37de56e9ed90e | **Public IPv4 address**<br>– | **Private IPv4 addresses**<br>10.1.3.6 |
| **IPv6 address**<br>– | **Instance state**<br>Running | **Public DNS**<br>– |
| **Hostname type**<br>IP name: ip-10-1-3-6.ec2.internal | **Private IP DNS name (IPv4 only)**<br>ip-10-1-3-6.ec2.internal | |
| **Answer private resource DNS name**<br>– | **Instance type**<br>t3.micro | **Elastic IP addresses**<br>– |

CloudShell | Feedback | Console Mobile App

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Inbox - Roy Baroudy (Stu | Step-by-step code explan | Course: EECE503Q - STD | Koh Lanta S03E02 - YouT | baroude | IAM | Global | Load balancers | EC2 | us | Two-Tier App

us-east-1.console.aws.amazon.com/ec2/home

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EC2 > Load balancers

- Spot Requests
- Savings Plans
- Reserved Instances
- Dedicated Hosts
- Capacity Reservations
- Capacity Manager New

**Images**
- AMIs
- AMI Catalog

**Elastic Block Store**
- Volumes
- Snapshots
- Lifecycle Manager

**Network & Security**
- Security Groups
- Elastic IPs
- Placement Groups
- Key Pairs
- Network Interfaces

**Load Balancing**
- Load Balancers
- Target Groups
- Trust Stores

**Auto Scaling**
- Auto Scaling Groups

Settings

## Load balancers (1/1) What's new?

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Actions | Create load balancer

Filter load balancers

| Name | State | Type | Scheme | IP address type | VPC ID | Availability Zones | Security groups |
|---|---|---|---|---|---|---|---|
| lab007-external-alb | Active | application | Internet-facing | IPv4 | vpc-004481cbe10308ffc | 2 Availability Zones | sg-0aa05e09647 |

## Load balancer: lab007-external-alb

Overview | Unhealthy target map | Show resource details

### lab007-external-alb

Last fetched seconds ago | Export

**Listeners (1)**
HTTP:80 — 1 rule

**Rules (1)**
Priority default — Forward to target group
Conditions (If)
*If no other rule applies*

**Target groups (1)** Info
Instance, HTTP — lab007-web-tg — 2 targets
2 | 0 | 0 | 0 | 0

**Targets (2)**
i-01c0d55c428585968 — Port 80 — Healthy
i-0a8df5f86a11d9169 — Port 80 — Healthy

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Inbox - Roy Baroudy (St | Step-by-step code expla | Course: EECE503Q - ST: | Koh Lanta S03E02 - Yo | baroude | IAM | Global | Security groups | EC2 | vpcs | VPC | us-east-1 | Two-Tier App

us-east-1.console.aws.amazon.com/vpcconsole/home

aws | Search [Alt+S] | United States (N. Virginia) | RoyBaroudy (8380-5432-1169) | RoyBaroudy

VPC > Your VPCs

**VPC dashboard**

AWS Global View

Filter by VPC

- **Virtual private cloud**
  - Your VPCs
  - Subnets
  - Route tables
  - Internet gateways
  - Egress-only internet gateways
  - Carrier gateways
  - DHCP option sets
  - Elastic IPs
  - Managed prefix lists
  - NAT gateways
  - Peering connections
  - Route servers
- **Security**
  - Network ACLs
  - Security groups
- **PrivateLink and Lattice**
  - Getting started
  - Endpoints
  - Endpoint services
  - Service networks
  - Lattice services
  - Resource configurations
  - Resource gateways
  - Target groups
  - Domain verifications New
- **DNS firewall**

## Your VPCs

VPCs | VPC encryption controls

**Your VPCs (1/2)** Info

Last updated less than a minute ago | Actions | Create VPC

Find VPCs by attribute or tag

| Name | VPC ID | State | Encryption c... | Encryption control ... | Block Public... | IPv4 CIDR | IPv6 CIDR |
|---|---|---|---|---|---|---|---|
| – | vpc-01aef5611c0916af9 | Available | – | – | Off | 172.31.0.0/16 | – |
| lab007-vpc | vpc-004481cbe10308ffc | Available | – | – | Off | 10.1.0.0/16 | – |

**vpc-004481cbe10308ffc / lab007-vpc**

Details | Resource map | CIDRs | Flow logs | Tags | Integrations

### Resource map Info

Show all details

**VPC**
Your AWS virtual network
- lab007-vpc

**Subnets (4)**
Subnets within this VPC
- us-east-1a
  - lab007-public-1
  - lab007-private-1
- us-east-1b
  - lab007-public-2
  - lab007-private-2

**Route tables (3)**
Route network traffic to resources
- Lab007-Private-RT
- Lab007-Public-RT
- rtb-0f9c94a1d574fa669

**Network Connections (2)**
Connections to other networks
- lab007-igw
- Lab-NAT-GW

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Inbox - Roy Baroudy (St | Step-by-step code expla | Course: EECE503Q - ST: | Koh Lanta S03E02 - Yo | baroude | IAM | Global | Security groups | EC2 | u | My parameters - AWS Sys | Two-Tier App

us-east-1.console.aws.amazon.com/systems-manager/parameters/

aws | Search [Alt+S] | United States (N. Virginia) | RoyBaroudy (8380-5432-1169) | RoyBaroudy

AWS Systems Manager > Parameter Store

# Parameter Store

My parameters | Public parameters | Settings

**You have upgrades available.** Manage settings to increase your capacity. Manage settings

## My parameters

View details | Edit | Delete | Create parameter

Search

1

| Name | Tier | Type | Last modified |
|---|---|---|---|
| /lab007/app_env | Standard | String | Wed, 18 Mar 2026 10:56:28 GMT |
| /lab007/db_password | Standard | SecureString | Wed, 18 Mar 2026 10:56:28 GMT |

CloudShell | Feedback | Console Mobile App

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ec2-user@ip-10-1-4-183:~ | Windows PowerShell

```
[ec2-user@ip-10-1-4-183 ~]$ exit
logout
Connection to 10.1.4.183 closed.
[ec2-user@ip-10-1-1-97 ~]$ exit
logout
Connection to 98.80.200.209 closed.
(base) PS C:\Users\royba\downloads\terraform-lab> ssh -i my-ec2-us.pem ec2-user@98.80.200.209
       #_
   ~\_  ####_
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   Amazon Linux 2023 (ECS Optimized)
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'

For documentation, visit http://aws.amazon.com/documentation/ecs
Last login: Wed Mar 18 11:14:17 2026 from 185.132.132.57
[ec2-user@ip-10-1-1-97 ~]$ ssh -i ~/.ssh/my-ec2-us.pem ec2-user@10.1.4.183
       #_
   ~\_  ####_
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   Amazon Linux 2023 (ECS Optimized)
   ~~       V~' '->
    ~~~         /
      ~~._.   _/
         _/ _/
       _/m/'

For documentation, visit http://aws.amazon.com/documentation/ecs
Last login: Wed Mar 18 11:34:13 2026 from 10.1.1.97
[ec2-user@ip-10-1-4-183 ~]$ curl http://localhost:3000/api/health
{"status":"healthy","tier":"backend","instanceId":"i-08da3d61cc0682b88","availabilityZone":"us-east-1b","timestamp":"2026-03-18T11:35:14.184Z"}[ec2-user@ip-10-1-4-183 ~]$
```

File Edit Selection View Go Run Terminal Help

503Q-lab7

EXPLORER

- 503Q-LAB7
  - .terraform
  - userdata
    - backend.sh
    - web.sh M
  - .gitignore
  - .terraform.loc... U
  - alb.tf
  - compute.tf
  - main.tf
  - network.tf
  - outputs.tf
  - secrets.tf
  - security.tf
  - terraform.tfstate
  - terraform.tfstate.b...
  - terraform.tfvars
  - variables.tf

> OUTLINE

> TIMELINE

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL AZURE

```
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 03m40s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 02m30s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 03m50s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 02m40s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 04m00s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 02m50s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 04m10s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 03m00s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 04m20s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 03m10s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 04m30s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 03m20s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 04m40s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 03m30s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 04m50s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 03m40s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 05m00s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 03m50s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 05m10s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 04m00s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 05m20s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 04m10s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 05m30s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 04m20s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 05m40s elapsed]
aws_internet_gateway.main: Still destroying... [id=igw-06da91d0188c4208d, 04m30s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 05m50s elapsed]
aws_internet_gateway.main: Destruction complete after 4m38s
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 06m00s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 06m10s elapsed]
aws_autoscaling_group.web: Still destroying... [id=lab007-web-asg, 06m20s elapsed]
aws_autoscaling_group.web: Destruction complete after 6m25s
aws_subnet.public_1: Destroying... [id=subnet-0f9559156fbbc24986]
aws_lb_target_group.web: Destroying... [id=arn:aws:elasticloadbalancing:us-east-1:838054321169:targetgroup/lab007-web-tg/20465dccbafa6b11]
aws_subnet.public_2: Destroying... [id=subnet-01bca6619108f489f]
aws_launch_template.web: Destroying... [id=lt-09ae96ba6ec7b1bd8]
aws_lb_target_group.web: Destruction complete after 1s
aws_launch_template.web: Destruction complete after 1s
aws_security_group.web: Destroying... [id=sg-04f126dcd2abfbced]
aws_iam_instance_profile.ec2_ssm: Destroying... [id=lab007-ec2-ssm-profile]
aws_subnet.public_2: Destruction complete after 1s
aws_subnet.public_1: Destruction complete after 1s
aws_security_group.web: Destruction complete after 2s
aws_security_group.alb: Destroying... [id=sg-0aa05e096472c22a8]
aws_security_group.bastion: Destroying... [id=sg-06c1e59df64a15f43]
aws_iam_instance_profile.ec2_ssm: Destruction complete after 2s
aws_iam_role.ec2_ssm: Destroying... [id=lab007-ec2-ssm-role]
aws_security_group.alb: Destruction complete after 1s
aws_iam_role.ec2_ssm: Destruction complete after 1s
aws_security_group.bastion: Destruction complete after 1s
aws_vpc.main: Destroying... [id=vpc-0044481cbe10308ffc]
aws_vpc.main: Destruction complete after 2s

Destroy complete! Resources: 32 destroyed.
(base) PS C:\Users\royba\Downloads\503Q-lab7>
```

powershell

powershell

main* ⊗ 0 ⚠ 0 [Azurite Table Service] [Azurite Queue Service] [Azurite Blob Service] Roy Baroudy (37 minutes ago) Ln 133, Col 23 Spaces: 2 UTF-8 LF Shell Script Go Live Prettier